



CORAL REEFS AND ATOLLS

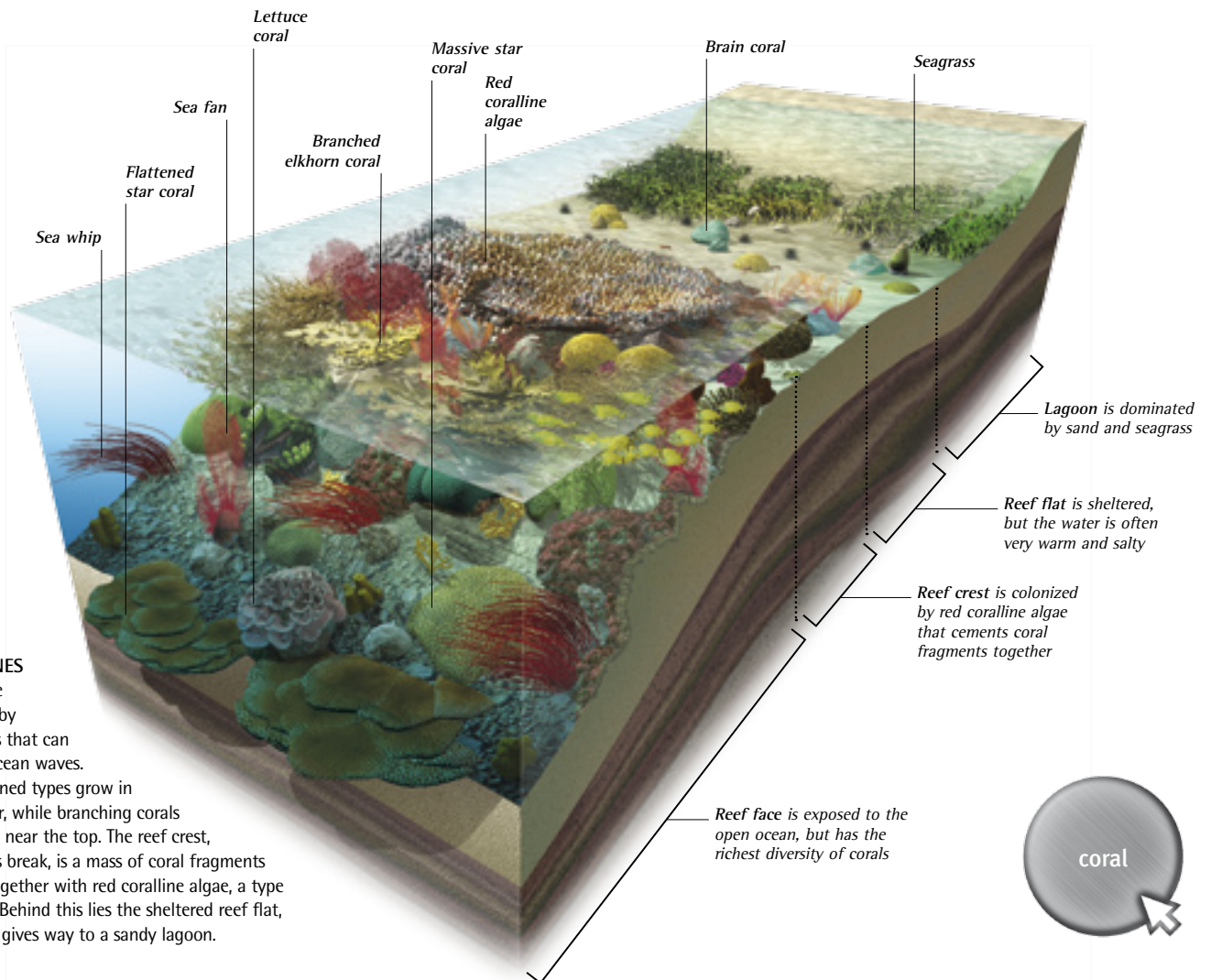
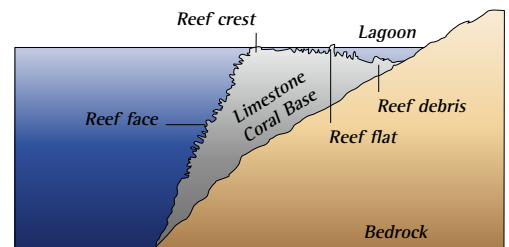
The clear waters of tropical oceans contain few of the nutrients that are essential to oceanic life, yet they support one of the richest ecosystems on Earth—tropical coral reefs. Their living wealth is based on a partnership between simple animals called corals and microscopic, plantlike organisms that make food using the energy of sunlight, just like phytoplankton. The growing corals form massive living reefs, populated by a dazzling diversity of animal life.

▲ REEF CORALS

Corals are animals that are closely related to sea anemones. They have the same cylindrical bodies crowned with stinging tentacles for catching small animals. Many live in interconnected colonies, including tropical reef-building corals, which are unique because they have microscopic plantlike zooxanthellae living in their tissues. These are supplied with vital nutrients by their hosts, and in return they use solar energy to make food that the corals need.

BUILDING REEFS ►

Since they need sunlight, reef corals live in clear, shallow water. Their soft bodies are supported by skeletons of limestone, which build up into rocky reefs. These reefs have steep sides facing the open ocean, but often enclose shallow lagoons.



► REEF ZONES

The reef face is colonized by strong corals that can withstand ocean waves. Broad, flattened types grow in deeper water, while branching corals are common near the top. The reef crest, where waves break, is a mass of coral fragments cemented together with red coralline algae, a type of seaweed. Behind this lies the sheltered reef flat, which often gives way to a sandy lagoon.